Two Forms of a Linear Equation

Standard form to slope-intercept form, back again, and the intercepts in between

Directions

- Standard form is $Ax + By = C$; slope-intercept form is $y = mx + b$. The two describe the same relationship.
- To go from standard to slope-intercept, isolate y: subtract the x-term from both sides, then divide *every* term by the coefficient of y.
- To go the other way, clear the fraction and move the x-term to the left so both variables sit on one side.
- Keep the quantities and units the story gives you attached to your answer.

1. A school spends \$36 on supplies, buying x notebooks at \$3 each and y pens at \$2 each: $3x + 2y = 36$. Rewrite the relationship in slope-intercept form, solved for y.

Answer: ________________

2. A club sells x child tickets at \$4 and y adult tickets at \$5, taking \$40 in all: $4x+5y = 40$. Rewrite the relationship in slope-intercept form, solved for y.

Answer: ________________

3. A caterer's plan for x trays of fruit and y trays of bread is $y = -\frac{2}{3}x + 8$. Rewrite it in the standard form $2x + 3y = C$ by finding the constant C.

Answer: ________________

4. A printing budget of \$60 buys x posters at \$5 each and y flyers at \$4 each: $5x + 4y = 60$.

(a) Find the x-intercept: how many posters if no flyers are bought?

Answer: ________________

(b) Find the y-intercept: how many flyers if no posters are bought?

Answer: ________________

5. A fundraiser's profit relationship is $2x - y = 7$, where x is the number of boxes sold and y is the profit in dollars. Rewrite it in slope-intercept form, solved for y.

Answer: ________________

6. A hiking club's plan for x hours of walking and y hours of rest is $y = \frac{3}{4}x - 5$. Rewrite it in the standard form $3x - 4y = C$ by finding the constant C.

Answer: ________________

7. A community fund of \$18 pays for x packets of seeds at \$6 each and y bags of soil at \$3 each: $6x + 3y = 18$.

(a) Find the x-intercept.

Answer: ________________

(b) Find the y-intercept.

Answer: ________________

(c) Use the two intercept points to find the slope of the line.

Answer: ________________

8. A booth takes exactly \$48 selling x small drinks and y large drinks, which is the relationship $3x + 4y = 48$. Some combinations are listed below.

Small drinks x	0	4	8	12
Large drinks y	12	9	6	3

(a) By how much does the number of large drinks change as x goes from 0 to 12?

Answer: ________________

(b) Rewrite $3x + 4y = 48$ in slope-intercept form, solved for y.

Answer: ________________

9. A paint mixture uses x litres of blue and y litres of white according to $y = -\frac{5}{2}x + 7$. Rewrite it in the standard form $5x + 2y = C$ by finding the constant C.

Answer: ________________

10. A community garden has \$120 to spend, with tomato plants at \$8 each and pepper plants at \$6 each: $8x + 6y = 120$, where x is the number of tomato plants and y the number of pepper plants.

(a) Rewrite the equation in slope-intercept form, solved for y.

Answer: ____________________

(b) Explain what the slope you found means for the gardener choosing between the two kinds of plant.

Answer: ____________________